



Virtual reality applications in computer science education: scoping review

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ABSTRACT

This scoping review explores the application of virtual reality (VR) technologies to computer science (CS) education. With the increasing scholarly interest in immersive and interactive learning, VR exhibits a strong potential in helping learners understand abstract CS concepts such as algorithms, object-oriented programming (OOP), and computational thinking.

This study conducted a comprehensive search across six databases and screened 3178 articles. A total of 30 peer-reviewed studies were included based on defined eligibility criteria. These studies were analyzed in terms of application features, educational focus, evaluation outcomes, and key findings.

The findings revealed that the majority of VR applications support conceptual learning instead of practical programming. Common design features included visual metaphors, game-inspired tasks, virtual panels, and embodied interactions, frequently tailored to specific CS topics or user groups. While numerous studies reported positive outcomes in learner performance and engagement, a number of challenges persist such as limited features for collaboration, usability issues, and lack of theoretical foundation. Gamification and constructivist approaches were frequently adopted, although without clear instructional grounding in certain cases.

This review addresses critical gaps in the existing literature by systematically analyzing VR design features and their reported associations with learning outcomes in CS education, identifying current trends, and persistent limitations. It offers insights for educators and developers on effective application designs and offers directions for future work, including broader content coverage, enhanced collaborative features, and increased integration of theoretically informed design strategies.

1. Introduction

Virtual reality (VR) has become a powerful tool in the domain of education, offering immersive, interactive, and engaging environments for learners aligned with the goals of Education 4.0 (Lampropoulos & Kinshuk, 2024). In recent years, its application to computer science (CS) education has garnered increased scholarly interest. Abad-Segura et al. (2020) emphasized that CS is the most popular keyword in the research on the sustainability of VR in higher education, with more than 70% of reviewed papers related to CS.

Rooted in CS, VR provides users with interactive and perceptually rich environments. These core strengths, including interaction, immersion, immediacy, involvement, and presence, have supported applications in various fields such as healthcare, entertainment, and education (Lampropoulos & Kinshuk, 2024). As education increasingly adapts to digital demands, students are seeking more engaging, interactive,

flexible, and meaningful learning experiences, which VR is well-suited to provide.

CS education frequently involves abstract and complex concepts, and VR demonstrates particular promise in this context. It enables learners to interact with content in embodied, visual, and playful ways, supporting the formation of mental models through spatial metaphors and simulations (Pirker et al., 2020). Given VR's promising potential, an increasing number of empirical studies have explored its application in CS education. For example, Pirker et al. (2020) compared student performances and preferences between a traditional 2D web-based interface and a VR environment delivered via High Tech Computer Corporation (HTC) VIVE for learning sorting algorithms. The results demonstrated that the majority of participants favored the VR environment and were inclined to use it for CS learning due to its heightened interactivity and enjoyment. Nevertheless, the study cited the small sample size, wide age gap (more than 15 years), and the diverse

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backgrounds of the participants as its limitations and called for further precise, large-scale research on the use of VR in CS education.

Although many studies have developed VR applications and evaluated their impacts on CS education, a systematic comparison across design features, such as learning scenarios, education topics, and devices remains lacking. Many studies have focused on technical implementation or user engagement but seldom examined whether these tools align with pedagogical theories or support specific learning outcomes. As a result, a disconnect exists between practical VR interventions and theoretical frameworks that guide effective instructional design.

In addition, several review studies identified papers related to VR applications in CS education. For example, Pirker et al. (2020) summarized topics, technologies, and target users from 13 articles between 2013 and 2019. Agbo, Sanusi, et al. (2021) conducted bibliometric and content analyses on 39 articles published from 2011 to 2020, mainly mapping methods and data types. Although these two reviews explored concepts and scenarios related to VR applications in CS education, detailed analyses of user performance and feedback remain limited. Recently, Lampropoulos and Kinshuk (2024) examined 112 gamified VR applications in CS education, providing insights into device usage, platform types, and elements of gamification. However, the work lacked emphasis on learning theories, features of application design, or educational impacts. Similarly, Al-Ansi et al. (2023) identified general VR research trends in education over the past decade, highlighting rapid growth but also highlighting persistent challenges such as limited scalability and high implementation costs. In general, studies that provided an integrated view of the design of these applications, learners' interaction with them, and their impacts on learning are scarce.

This scoping review aims to fill these research gaps by providing a comprehensive synthesis of peer-reviewed studies on VR applications in CS education. It explores which categories of CS topics are taught through VR, how applications are designed and grounded in relevant learning theories, which technologies and interaction methods are employed, and what outcomes are reported. By mapping current design practices to learning outcomes and theoretical foundations, this review offers actionable insights for educators, researchers, and developers who aim to improve VR use in CS education.

1.1. Objective

To address the abovementioned research gaps, the study poses the following research questions.

(RQ1) What is the context of VR technologies and interventions in CS education?

(RQ2) What are the key technologies, theoretical foundations, and design features of VR application in CS education?

(RQ3) How do VR applications influence learning outcomes in CS education?

By reviewing and analyzing existing applications, this study aims to provide practical guidelines for the design and evaluation of VR tools in CS education. The results are expected to highlight the current development status and discuss the design features, effectiveness, scope of use, and theories of VR instructional applications.

2. Methods

This scoping review follows the methodological framework proposed by Arksey and Malley (2005), with the objective of systematically identifying and examining studies that have developed and evaluated VR applications within CS education, particularly in programming-related contexts. To address the research questions, a comprehensive search strategy was designed, clearly defined eligibility criteria were applied, and a structured screening and extraction process was followed.

2.1. Search sources

A broad search was conducted across six major bibliographic databases—Google Scholar, SpringerLink, Scopus, IEEE Xplore, Association for Computing Machinery Digital Library, and Web of Science—due to their extensive coverage of CS, technology-enhanced learning, and VR research. Although similar keywords were used across platforms, search strings were tailored to the functionality and indexing structures of each database. All searches were conducted on April 3, 2025.

2.2. Search strategy

The study developed a conceptual categorization of search terms to guide consistent query construction given that VR and CS education span multiple overlapping domains. Three major term groups were defined, namely, VR-related terms (e.g., “virtual reality” and VR), CS-related terms (e.g., computer, computing, programming, software, algorithms, information technology, information systems, and data science), and education-related terms (e.g., learning, instruction, pedagogy, curriculum, and assessment). These keywords were selected due to their prevalence in the literature on CS education and VR-based learning environments. Boolean combinations that incorporated synonyms and derivatives were systematically adjusted for each database.

To promote transparency and reproducibility, the following text provides representative search queries. For example, Google Scholar queries used the *intitle* operator to increase specificity, which resulted in combinations such as (“virtual reality” OR VR) AND intitle:(computer OR computational OR computing OR algorithm OR software OR program OR programming OR coding OR “information technology” OR “information systems” OR “data science”) AND intitle:(learning OR education OR instructional OR pedagogy OR curriculum OR assessment). Comparable title-field queries were used in SpringerLink, Web of Science, IEEE Xplore, and Scopus, with slight modifications to accommodate database-specific syntax. These queries ensured broad coverage while maintaining relevance to VR in CS education.

2.3. Eligibility criteria

To ensure that the included studies were contemporary, relevant, and methodologically robust, the study applied several eligibility criteria during screening. First, the review was limited to studies published between 2018 and 2025, thus reflecting recent developments in VR hardware, interaction techniques, and educational applications. Only studies written in English and published in peer-reviewed venues were considered.

To be eligible, studies needed to focus explicitly on CS education, which for the purposes of this review included programming, computational thinking, algorithms, and computer architecture. Studies unrelated to CS education (e.g., general robotics and engineering) were excluded. Furthermore, the reviewed interventions needed to include an immersive or semi-immersive VR component with a visual interface. Each study was required to report user evaluations—whether qualitative, quantitative, or mixed methods—to enable meaningful assessments of educational impacts. To enhance specificity during screening, essential terms such as virtual reality and computer science were required to appear in the title or abstract.

2.4. Study selection

Study selection was conducted using Covidence and followed three sequential stages. First, Covidence automatically identified and omitted duplicate records generated from the combined database search results. Second, two reviewers independently screened the titles and abstracts of the remaining publications according to the abovementioned eligibility criteria. Covidence flagged conflicting decisions, thus prompting reviewers to provide written justifications prior to engagement in a

structured discussion to resolve discrepancies. Third, both reviewers then retrieved and independently reviewed full-text articles. Ambiguous or borderline cases, especially those referring broadly to computing without explicitly addressing CS education, were evaluated on a case-by-case basis. Only studies that met all inclusion requirements and presented sufficient educational data were retained for synthesis.

2.5. Data extraction

Data extraction followed a structured template developed within Covidence. For each study, the researchers recorded bibliographic information (e.g., authors, title, and year of publication) and characteristics relevant to the research pathway, including whether a VR application was part of a long-term or an iterative project. We extracted details on educational topics (e.g., programming, algorithms, and object-oriented concepts), participant demographics (e.g., age group, level of education, number of participants, and prior programming experience), features of VR applications such as device type (e.g., Oculus, HTC VIVE, and Google Cardboard), design or educational theories informing development, and characteristics of VR learning environments.

Evaluation-related data, including learning content, performance outcomes (e.g., task completion and test scores), and user feedback (e.g., usability, satisfaction, and acceptance), were also collected. Each study

was coded to determine whether it reported VR as an effective or beneficial tool for CS learning. Two reviewers independently conducted data extraction and thematic coding, grouping findings into categories aligned with the topics in the research questions: implementation context, application design, theoretical foundation, and educational outcome. Any inconsistencies were discussed until consensus was reached, thereby ensuring reliability and transparency in the analytical process.

3. Results

This section presents the main results. Section 3.1 provides an overview of the study selection process. Section 3.2 addresses RQ1 by examining the implementation of VR in CS education. Section 3.3 responds to RQ2 through an analysis of application technologies and design features. Finally, Section 3.4 presents evidence related to RQ3, including learning outcomes and user feedback.

3.1. Overview of search strategy and included studies

A total of 3178 records were initially retrieved, including 332 duplicates. After screening the titles and abstracts, 2749 records were excluded. The remaining 97 articles underwent full-text review, which resulted in 30 studies that met the inclusion criteria (Fig. 1). Two of

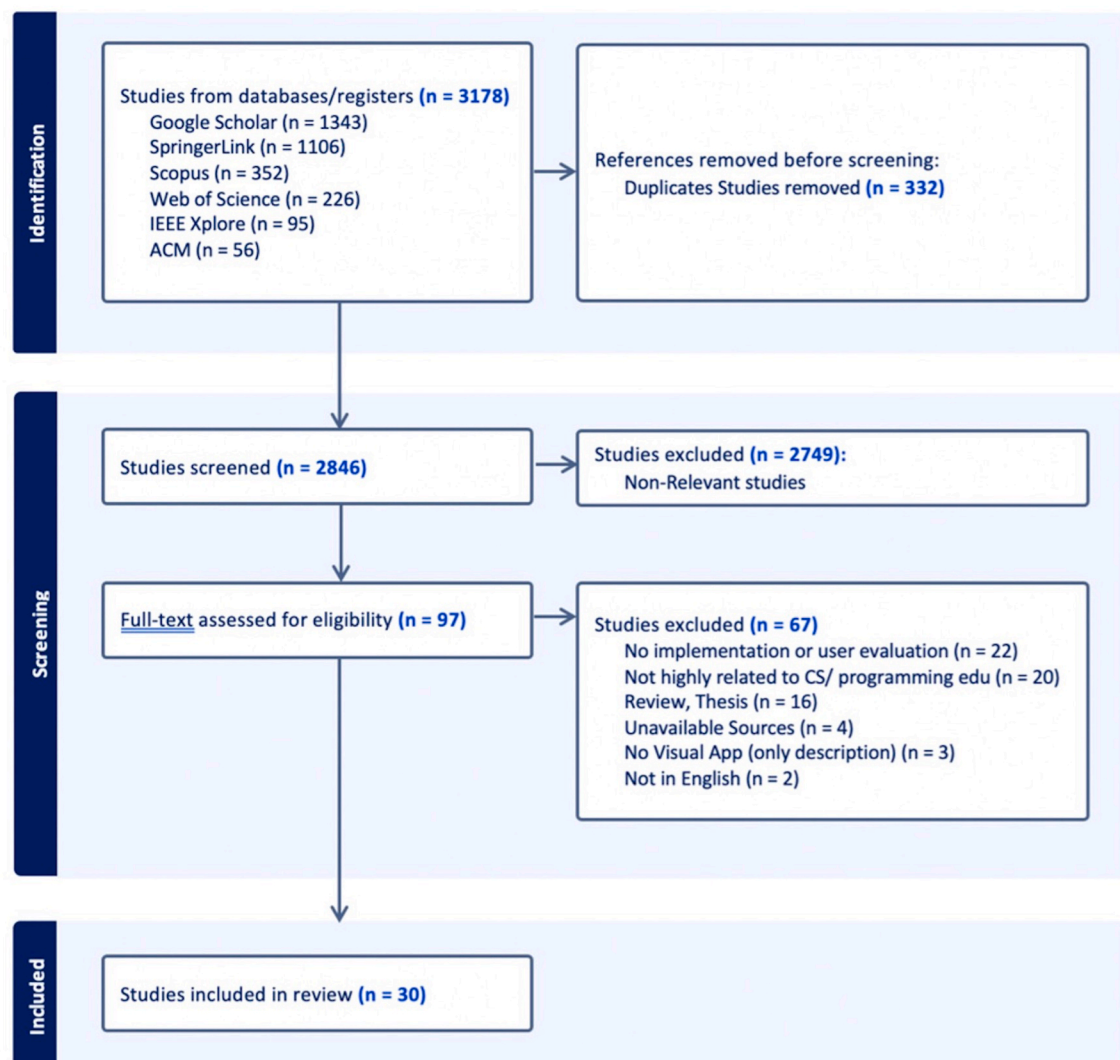


Fig. 1. PRISMA flowchart.

them described the same application, Imikode, thus yielding a final sample of 29 unique VR applications designed for CS education.

To enhance transparency, [Appendix A](#) provides a comprehensive list of the selected applications, including year of publication, topic category, and authors. In cases in which many papers reported on the same application as a part of long-term studies, only one representative study was included in the statistical summaries. For example, we selected two papers related to Imikode but retained only one in the statistical result. Moreover, nine applications were unnamed, which may reflect early-stage development.

Articles published between 2021 and 2024 comprise the majority of the sample (26/29, 90%), while no articles were published in 2020. A potential explanation is the impact of COVID-19, which disrupted research progress given that the majority of VR application studies required face-to-face testing. After the pandemic, researchers resumed testing and subsequently published their findings.

3.2. RQ1: contexts of VR implementation in CS education

To elucidate the contexts in which VR applications are implemented in CS education (RQ1), analysis included participant demographics, level of education, usage settings, subject areas, and research path.

3.2.1. Participant demographics

We collected data on participant demographics from the 29 selected articles; notably, researchers frequently recruited individuals with specific experiences or attributes tailored to the requirements of their experiments. The selected studies mainly discussed people with experience with various applications, many studies were conducted in multiple phases (e.g., preliminary surveys), and not all participants were included in the final counts. For instance, [Klochko et al. \(2022\)](#) conducted a Tower of Hanoi study and collected 1664 survey responses on VR use in CS education. Given the complexity of these study designs, the number of participants may be inconsistent or include duplicate entries. In this review, the demographic summary is based on the understanding of the researchers and may include certain degrees of estimation error. For precise data, readers should refer to the original papers.

All participants were classified as adults and children ([Fig. 2](#)). Only 9

out of the 29 selected articles recorded participants' age. For those that did not explicitly provide this report, categorization was inferred based on level of education. The majority of adult users were aged between 18 and 25 years, corresponding to the age of a typical college student. In contrast, the majority of child users were aged between 11 and 16 years, typically corresponding to middle school levels (grades 5–8).

3.2.2. Levels of education and settings

The reviewed VR applications were primarily implemented in the contexts of university and middle school education. University students constituted the largest group (837 out of 1337 participants across 21 studies), accounting for 62.6%. Approximately 30% (448 participants) were from grades 5 to 7, representing the middle school level. For the remaining 52 participants, 10 reported no prior programming experience ([Segura et al., 2020](#)), while 20 were professionals involved in a VR-based training study focused on computer hardware ([Fasihuddin, 2023](#)). [Fig. 3](#) depicts the distribution of participants in terms of level of education. VR applications were implemented in various educational settings, including in-class sessions, laboratory-based experiments, and remote or self-guided environments. Several studies adopted mobile or web-based VR platforms (e.g., [Akbulut et al., 2018](#); [Saleh et al., 2021](#); [Martins Nunes Avellar et al., 2024](#)) for remote or informal learning. While the majority of interventions were designed for individual use, a few incorporated collaborative features or teacher–student interactions (e.g., [Hui et al., 2024](#); [Lui et al., 2025](#)), which are further discussed in [Section 3.3](#).

3.2.3. Subject areas

The selected applications addressed a wide range of CS-related topics. Statistically, the most frequently addressed topics include algorithms ([Akbulut et al., 2018](#); [Alexander et al., 2022](#); [Dewi et al., 2024](#); [Klochko et al., 2022](#); [Pirker et al., 2021](#); [Yahui et al., 2024](#)), object-oriented programming (OOP) ([Lui et al., 2025](#); [Segura et al., 2020](#); [Sunday et al., 2022](#); [Wee et al., 2022](#)), and computational thinking ([Agbo, Olaleye, et al., 2023](#); [Li et al., 2024](#); [Parmar et al., 2023](#); [Sukirman et al., 2024](#)). Introductory programming concepts are also commonly explored, with three studies focusing on this area (i.e., [Ekman, Solsona, & Quintero, 2024a, 2024b](#); [Martins Nunes Avellar](#)

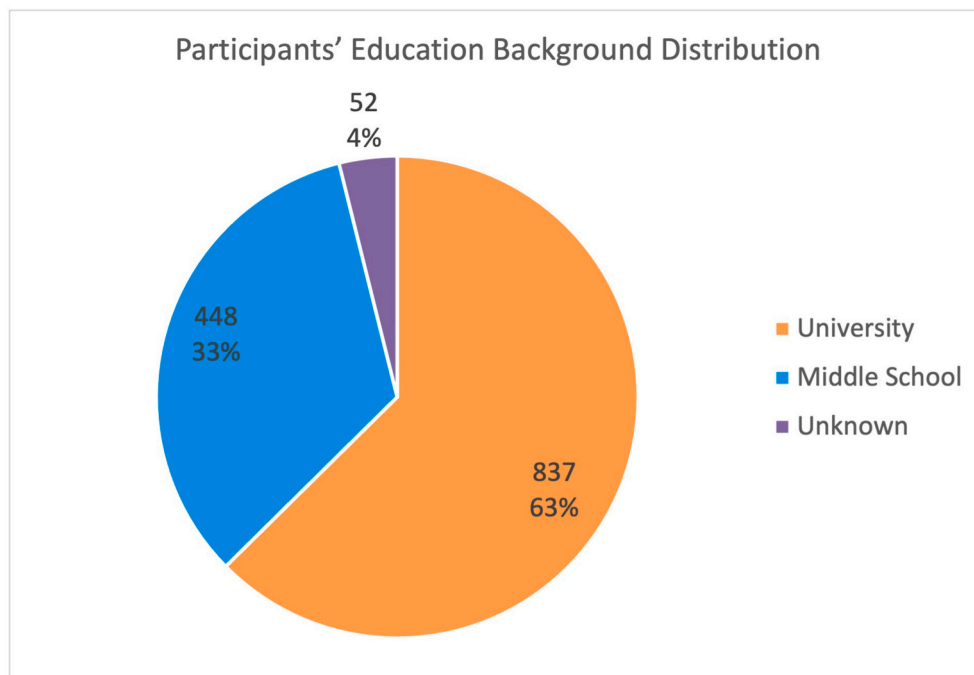


Fig. 2. Age distribution of participants.

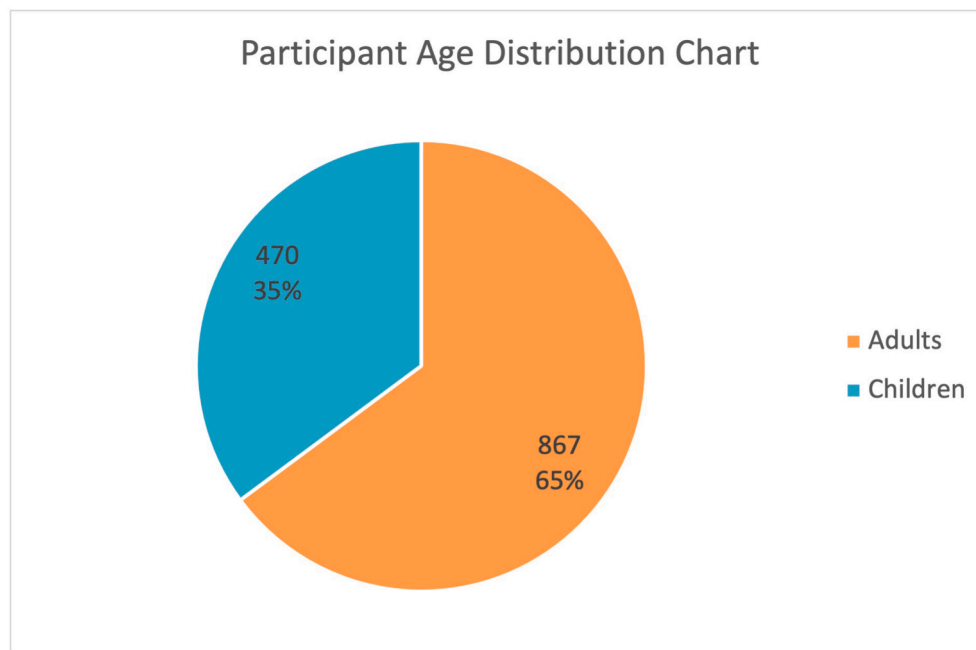


Fig. 3. Distribution of participants' level of education.

et al., 2024; Moersler et al., 2024).

Two studies developed applications for teaching Python (Saleh et al., 2021; Theethum et al., 2021), a language known for its readability and wide applicability (e.g., in data analysis). Another two studies addressed concepts in computer architecture (Hui et al., 2024; K. H. A. Tan et al., 2023), a foundational but abstract and complex topic in CS. In contrast to software-focused studies, two works applied VR to computer hardware education (Cheng-Hsiu Li, 2023; Fasihuddin, 2023), while one study incorporated Arduino-related hardware operations (Jin, Bai, Song, Yu, & Yue, 2024). Other studies explored specialized areas, including deep learning in AI (Hunvik et al., 2021), spatial reasoning (Lu et al., 2022), computer graphics (Heinemann & Schroeder, 2023), unity development skills (Anastasi & Munna, 2023), and drone programming (Jin, Bai, Song, Yu, & Yue, 2024).

3.2.4. Research path

A notable feature of the selected studies is the extent to which many VR applications emerge from long-term or iterative research programs instead of isolated, single-phase prototypes. Across several projects, researchers refined their applications across many years, expanding their pedagogical models, improving interface designs, and conducting repeated evaluations with increasingly diverse learner groups. These developmental trajectories provide important insight into the evolution of VR interventions within CS education and the progressive improvements to usability and learning outcomes made by research teams.

One group of sustained research programs focused on the development of VR tools for teaching algorithms, computational thinking, and introductory CS concepts. For example, building on earlier work grounded in the principles of CS Unplugged, Dengel (2020) demonstrated the usefulness of educational virtual environments for CS learning, which was supported by a prior systematic review conducted by Pirker et al. (2020) highlighting VR's educational potential. Later, Pirker et al. (2021) extended this line of inquiry by designing a new VR application for teaching algorithms and comparing between its VR and desktop variants. Similarly, the long-running Dancing Alice and Virtual Environment Interaction (VEnvI) projects showcase an iterative pedagogical model that integrates embodied interaction, dance, and programming instruction. The results of initial pilot studies on elementary students (Daily et al., 2014) informed subsequent iterations of the VEnvI

platform, which underwent multiple rounds of design improvement and testing on middle and secondary school learners (Leonard et al., 2021; Lin et al., 2017; Parmar et al., 2016). The series culminated in comparative studies that demonstrated the importance of immersion and avatar design for computational thinking development (Parmar et al., 2023). Another example is the Interactive Programming in Virtual Reality (iProgVR) project, initially introduced as a visualization-oriented framework for teaching array operations (Wee & Yap, 2021) and subsequently expanded through rigorous methodological refinements, including the addition of pre-post-testing and controlled comparisons (Wee et al., 2022).

A second group of iterative research programs emphasized game-based VR learning environments, particularly those addressing OOP and computational thinking. The Imikode project exemplifies this trajectory, beginning with the development of a prototype VR game implemented using Google Cardboard (Bouali et al., 2019). The research progressed to an applied intervention aimed to address high rates of failure in programming courses among university students in Nigeria (Sunday et al., 2020). Subsequent studies extended this work by examining the relationship between VR gameplay and the acquisition of OOP concepts (Sunday et al., 2022); the application was later enhanced through real-time feedback, intelligent agent support, and additional game levels (Sunday et al., 2023). A similar long-term progression characterizes the iThinkSmart platform. Early work by Agbo et al. (2019, 2021) explored the broad roles of computational thinking and VR in CS education through literature reviews and bibliometric analyses. Building on these foundations, the team developed increasingly sophisticated game-based learning environments, including VR versions tested using university CS students (Agbo, Oyelere, et al., 2021). The authors then examined the influence of gamification and immersion on learners' perceptions (Agbo, Olaleye, et al., 2023). Similarly, the My Reality (MYR) environment evolved from an A-Frame-based tool for creating virtual scenes (Berns et al., 2019) to a classroom-tested platform that demonstrated support for spatial visualization and conceptual understanding (Lu et al., 2022).

A third cluster includes projects that are still emerging but demonstrating promising iterative development. The Responsive Pixel Virtual Reality (RePiX) VR system, first introduced as a tool for teaching computer graphics (Heinemann et al., 2022), underwent several

refinements, including expanded sample sizes, direct comparisons with desktop counterparts, and the integration of the User eXperience in Immersive Virtual Environments (UXIVE) model along with the text-sorting technique for usability and assessment of user experiences (Heinemann et al., 2023; Heinemann & Schroeder, 2023). These studies highlighted the potential of VR to support learners with varying levels of prior knowledge while underscoring the importance of testing for systematic usability.

In summary, these long-term and iterative research programs illustrate the evolution of VR applications over time through cycles of design, testing, feedback, and theoretical refinement (Agbo, Olaleye, et al., 2023; Heinemann & Schroeder, 2023; Lu et al., 2022; Parmar et al., 2023; Pirker et al., 2021; Sunday et al., 2022; Wee et al., 2022). They demonstrated sustained scholarly engagement in VR-based CS education and provided valuable guidance for new research teams in the field. The evolution of these platforms from early prototypes to well-established educational tools demonstrates the development of implementation contexts, pedagogical strategies, and technological features in response to user needs and empirical evidence. In contrast, several other studies included in this review represent early-stage prototypes that address specific CS topics such as Python programming, data structures, algorithms, or computer hardware. Although these emerging studies presented innovative ideas and promising results, many require further development, larger sample sizes, and extended evaluation to establish broader educational applicability. Taken together, these research paths reflect not only the diversity of VR interventions in CS education but also the dynamic, iterative nature of their development.

3.3. RQ2: technologies, theoretical foundations and, design features of VR applications in CS education

To address RQ2, information on three key aspects of each VR application was extracted: (1) the technological components used, including VR toolkits, devices, and platforms, (2) design features of VR learning environments such as interaction styles, virtual scenarios, visual metaphors, and embedded game mechanics, and (3) theoretical or pedagogical foundations that guide their development.

Table 1 summarizes these elements by outlining each application's VR toolkits, scenario, design features, theoretical foundation, and design-related findings. The following subsections synthesize these results into broad themes regarding technologies and platforms, theoretical bases, interaction designs, content-driven patterns, and collaborative features.

3.3.1. VR devices and platforms

The choice of VR device is typically linked to the platform on which the application runs. Devices connected to computers tend to support complex and high-performance designs. Table 1 indicates that the majority of applications in CS education utilize advanced headsets such as the Oculus (Anastasi & Munna, 2023; Parmar et al., 2023; Theethum et al., 2021; Wee et al., 2022), HTC VIVE (Alexander et al., 2022; Jin, Bai, Song, Yu, & Yue, 2024; Pirker et al., 2021; Sunday et al., 2022), and various models from the Quest series. In recent years, standalone headsets, such as the Oculus Quest 2 (Ekman et al., 2024a, 2024b; Sukirman et al., 2024) and Meta Quest Pro (Lui et al., 2025), gained popularity due to their wireless functionality and user-friendly design.

Additionally, several instructional applications are mobile-based, including those in Akbulut et al. (2018), Cheng-Hsiu Li (2023), Martins Nunes Avellar et al. (2024), Saleh et al. (2021), and Sunday et al. (2022). Notably, one application (Cheng-Hsiu Li, 2023) is compatible with consoles, mobile devices, and web platforms. Among the selected studies, Google Cardboard is the most commonly used mobile VR device—valued due to its affordability, simplicity, and broad compatibility. Only Imikode utilizes a VR BOX headset paired with a Bluetooth controller (Sunday et al., 2022). The Tower of Hanoi application, developed using CoSpaces Edu, is web-based and does not require users

to wear specialized VR equipment (Klochko et al., 2022).

3.3.2. Theoretical foundations

The VR applications were informed, explicitly or implicitly, by a range of educational and cognitive theories relevant to immersive learning. While several studies directly referenced frameworks such as constructivism or gamification, many others embedded design choices that aligned with broad learning theories commonly used in VR research. Taken together, these foundations can be grouped into four overarching categories, namely, constructivist learning; gamification and game-based learning; embodied cognition; and situated or experiential learning.

Constructivism was the most frequently cited theoretical perspective (Akbulut et al., 2018; Cheng-Hsiu Li, 2023; Heinemann & Schroeder, 2023; Jin, Bai, Song, Yu, & Yue, 2024; Li et al., 2024; Lu et al., 2022; Saleh et al., 2021; Segura et al., 2020; Wee et al., 2022). These applications promoted knowledge accumulation among learners through active engagement with virtual contents. Examples include manipulating 3D scenes in MYR (Lu et al., 2022), experimenting with algorithms in VR-ENITE (Akbulut et al., 2018), and interacting with memory-drawer metaphors in iProgVR (Wee et al., 2022). Constructivist design was particularly evident when students could observe system changes, test hypotheses, and explore cause-and-effect relationships through interactive VR features.

Gamification was also widely embedded across the applications, typically through puzzles, levels, rewards, challenges, and interactive feedback loops (Alexander et al., 2022; Dewi et al., 2024; Hunvik et al., 2021; Klochko et al., 2022; Parmar et al., 2023; Sukirman et al., 2024; Sunday et al., 2022; Theethum et al., 2021). Many applications used game mechanics to enhance motivation. For example, CT Saber incorporated sword-swinging mechanics to teach shortest-path logic (Sukirman et al., 2024), while the Escape-Room VR game positioned programming tasks within a narrative problem-solving framework (Hui et al., 2024). These approaches reflected the principles of game-based learning, in which structured challenge designs supported engagement and scaffolded understanding.

Apart from these explicit theoretical references, many applications implicitly drew on embodied cognition, which posits that physical actions and spatial interactions can support cognitive processes. VR supports this naturally by enabling users to manipulate virtual objects as if they were physical. Embodied learning was evident in VEnvI, in which choreographing and observing dance movements supported computational thinking (Parmar et al., 2023); in Thinkercise, in which learners used body movements to select answers or avoid obstacles (Theethum et al., 2021); in iProgVR, which used spatial metaphors such as drawers to illustrate memory and array processing (Wee et al., 2022); and in RePiX VR, which required coordinated VR-controller actions during interactive rendering tasks (Heinemann & Schroeder, 2023). These applications fostered bodily engagement to help learners map abstract programming ideas onto physical actions.

Situated learning was also reflected in several designs, emphasizing knowledge acquisition within authentic or context-rich environments. VR's immersive capacities enabled learners to experience CS concepts in realistic scenarios such as controlling drones or smart-farm devices in DVRT and EduCodeVR (Jin, Bai, Song, Yu, & Yue, 2024; Jin, Bai, Song, Yu, & Yue, 2024); exploring sensors and hardware in the Arduino Day virtual museum (Anastasi & Munna, 2023); or solving binary, ASCII, and assembly puzzles in the Escape-Room VR game (Hui et al., 2024). In relation, experiential learning principles were evident in applications that relied on iterative practice, immediate feedback, and repeated experimentation to refine understanding.

Finally, several studies were implicitly based on multimedia learning principles, including dual-coding, modality effects, and visual-spatial reasoning. For example, the MYR platform emphasized the spatial construction of code-driven scenes (Lu et al., 2022), while RePiX VR incorporated multimodal cues through graphics, avatars, and interactive

Table 1

VR toolkits, scenarios, design features, and findings.

No.	Application Name	VR Toolkits	VR Scenario	Design Features	Theoretical Foundation	Findings
1	VR-ENITE (Akbulut et al., 2018)	Google Cardboard iOS smartphones	Virtual classroom; students view 3D sorting algorithm animations; a tutor controls the screen via a tablet	3D algorithm animations; instructor-controlled display; low-cost mobile setup	Constructivism	1. Supported traditional teaching in software engineering. 2. Feasible with simple VR glasses, instead of full head-mounted displays (HMDs)
2	VR-OCKS (Segura et al., 2020)	HTC VIVE SteamVR	Puzzle-based navigation in which users placed blocks as program instructions	Spatial programming blocks; progressive puzzle levels; embodied navigation	Gamification; embodied cognition	1. Children learned controllers faster than expected. 2. Learners rushed to the next level instead of checking answers.
3	Unnamed (Hunvik et al., 2021)	Oculus Quest	Escape-room scenes representing deep learning concepts	Themed rooms as neural network layers; 3D cube insertion; physical puzzle mechanics	Gamification; situated learning	1. Escape-game format increased engagement. 2. Excessive text decreased usability; more audio/video recommended.
4	Unnamed (Pirker et al., 2021)	HTC VIVE	Multiple displays with draggable algorithm steps and adjustable speed	Drag-and-drop sequencing; speed control; visual feedback	Gamification	1. Increased engagement/emotion but not necessarily learning. 2. Web version was better for large cohorts; VR for small groups.
5	Unnamed (Saleh et al., 2021)	Google Cardboard Mobile VR	Python-based maze escape with tutorial videos; explanations are displayed when incorrect decisions are made	Maze metaphor for decision-making; branching scenarios; instructional pop-ups	Constructivism; gamification	1. Experts noted low-quality graphics. 2. Scenarios lacked richness for deep learning. 3. Beginners needed more time to understand syntax.
6	Thinkercise (Theethum et al., 2021)	Oculus Rift	Panel-based Python questions; users avoid obstacles while selecting cubes	Physical movement integrated with quizzes; calorie-tracking and difficulty settings	Gamification; embodied cognition	Encouraged physical activity; adaptable difficulty supported engagement.
7	VEnvI (Parmar et al., 2023)	Oculus Rift Kinect v2	Avatar choreography scenes linked to CS concepts	Embodied dance programming; first-person perspective; motion-capture	Embodied cognition; gamification	Ease of use, dancing interaction, and code visualization improved learner responses.
8	CSVIL (Alexander et al., 2022)	HTC VIVE	Classroom modules with tutorials, games, and virtual lecture theaters	Multi-module environment; instructor–student interaction features; scripted tutorials	Gamification; situated learning	VR version more immersive; equipment heavy and expensive.
9	Imikode (Sunday et al., 2022)	VR BOX + Bluetooth controller	Non-player character (NPC)-guided object-oriented programming (OOP) tasks (variables, objects, and instantiation)	NPC scaffolding; interactive OOP panels; smartphone-based HMD	Gamification	VR caused dizziness; issues with visual impairments and safety concerns.
10	iProgVR (Wee et al., 2022)	Oculus Rift	Array elements as drawers; users manipulate loops and values	Drawer-based memory metaphor; embodied manipulation of array indices	Constructivism; embodied cognition	Fun and intuitive; other users reported VR sickness and physical demand.
11	Tower of Hanoi (Klochko et al., 2022)	CoSpaces Edu (web VR)	Users move plates between rods in a classic puzzle	Drag-and-drop tower mechanics; minimal VR requirements	Gamification	–
12	My Reality (MYR; Lu et al., 2022)	Headsets, phone, and cardboards	Creative scene building with JavaScript	Code-driven 3D object manipulation; project-sharing templates	Constructivism	Students shared projects and explained code; improved spatial visualization.
13	iThinkSmart (Agbo, Olaleye, et al., 2023)	HMDs + smartphone controllers	Mini-games: River Crossing, Treasure Hunt, and Tower of Hanoi	Puzzle-based logic tasks; clear goals; progressive challenge	Gamification	1. Goal clarity strongly influenced comprehension. 2. Immersion/interaction exerted indirect effects.
14	Unnamed (Cheng-Hsiu Li, 2023)	Google Cardboard Android	Hardware assembly tasks	Virtual hardware components; assembly mechanics; drag-and-drop	Constructivism; situated learning	Knowledge retention for a long term.
15	Unnamed (Fasihuddin, 2023)	VR headset + controller	Hardware setup on virtual table	Point-and-place interactions; component wiring; Technology Acceptance Model (TAM)-based evaluation	Situated learning; experiential learning	High acceptance measured using TAM.
16	RePiX VR (Heinemann & Schroeder, 2023)	VR headset	Guided tour led by a robotic avatar; computer graphics tasks	Avatar-guided steps; visualization of rendering pipeline; task-based scenarios	Constructivism; multimedia learning	UXIVE model and text-sorting technique demonstrated strong improvements in usability.

(continued on next page)

Table 1 (continued)

No.	Application Name	VR Toolkits	VR Scenario	Design Features	Theoretical Foundation	Findings
17	N (K.H.A. Tan et al., 2023)	HMD + Desktop VR	CPU simulation rooms: tutorials, assignments, quizzes	Drag-and-drop CPU components; interrupt handling; sequential room progression	Gamification; multimedia learning	Improved understanding of architecture concepts; better interrupt-related performance.
18	CT Saber (Sukirman et al., 2024)	Oculus Quest 2	Sword-based maze navigation for shortest-path tasks	Rhythmic slashing mechanic; tile-based challenges; obstacle avoidance	Gamification; embodied cognition	Perceived usefulness predicted satisfaction; strong acceptance.
19	Z3VR (Moersler et al., 2024)	HMD	Historical Z3 computer simulation	Button panels; programming board; film-punching metaphor	Situated learning; experiential learning	Effective for novices; good knowledge transfer.
20	Escape-Room VR (Hui et al., 2024)	Oculus	Sci-fi escape challenges requiring binary, ASCII, and assembly tasks	Adaptive virtual assistant; timed puzzles; narrative hints	Gamification; situated learning	High immersion; curiosity agent improved engagement.
21	Unnamed (Yahui et al., 2024)	HMD/Desktop	3D visualization of DS/Algo (arrays, stacks, and graphs)	Auto-demo + interactive exercises; pathfinding tasks	Constructivism; multimedia learning	Improved learning of architecture concepts; higher levels of motivation.
22	Academic Museum of Technology (Anastasi & Munna, 2023)	Oculus Rift	Virtual museum + Arduino treasure hunt	Retro tech objects; sensor-based interactions; narrative tasks	Situated learning; peer learning (suitable for experiential/situated learning)	AR/VR increased motivation and excitement.
23	Unnamed (Li et al., 2024)	HMD	3D environment design and problem solving	Scene construction; logic-based interaction; creative coding tasks	Constructivism; experiential learning	Significant gains in CT across five dimensions.
24	Codezeum (Ekman et al., 2024a, 2024b)	Meta Quest 2	Gallery of programming puzzles	Node-link programming; logic gates; flow manipulation	Gamification	VR is more engaging; desktop is more efficient.
25	EduCodeVR (Jin, Bai, Song, Yu, & Yue, 2024)	Pico 4	Smart farm and traffic simulation	Block-based programming; first-person device control; environmental effects	Constructivism; situated learning	VR improved STEAM literacy; strong gender differences; novelty faded.
26	SSPOT-VR (Martins Nunes Avellar et al., 2024)	Google Cardboard	Robot programming in a space station	Block-based command assembly; virtual tutor; level progression	Constructivism; experiential learning	Significant knowledge gains; highly accessible; strong retention.
27	Unnamed (Dewi et al., 2024)	VR headset + controller	Algorithm and math puzzle rooms	Object interaction tasks; code panels; quiz-based review	Gamification; experiential learning	VR improved outcomes and supported diverse learning styles.
28	DVRT (Jin, Bai, Song, Yu, & Yue, 2024)	HTC VIVE + drone	Smart farm drone-control tasks	Drag-and-drop drone logic; real-world sensor integration	Gamification; situated learning	Strong improvements in CT and STEM literacy.
29	VRLOOP (Lui et al., 2025)	Meta Quest Pro	Shared VR room for collaborative OOP tasks	Multiuser interactions; object manipulation; co-op learning puzzles	Constructivism; multimedia learning; collaborative learning (aligned with situated/experiential learning)	Enhanced OOP understanding; collaboration improved learning quality.

panels (Heinemann & Schroeder, 2023). These design choices aligned with Mayer's cognitive theory of multimedia learning, even when the authors did not explicitly cite it (Lui et al., 2025).

3.3.3. Design features

Beyond technological and theoretical considerations, the reviewed studies reveal a range of design strategies that shape how learners engage with CS concepts in immersive environments. To systematically analyze these strategies, design features are examined across three interrelated dimensions: (1) interaction and scene design, (2) content-driven interaction patterns, and (3) collaborative and multiuser design.

Interaction and scene design refers to the visual, spatial, and interface elements that structure user engagement within the virtual environment. Content-driven interaction patterns capture recurring design approaches that are shaped by specific CS topics, such as algorithms or object-oriented programming. Collaborative and multiuser design addresses the extent to which VR applications support peer interaction, teacher–student engagement, or cooperative problem solving.

This three-part categorization provides a coherent analytical lens for understanding how pedagogical intentions are translated into immersive learning experiences.

3.3.3.1. Interaction and scene design. The researchers designed different virtual scenarios (Fig. 4) but shared a number of interesting features as follows.

First, the majority of applications incorporated virtual displays

within their environments to guide user interaction and provide textual instructions ((Agbo, Olaleye, et al., 2023; Alexander et al., 2022; Dewi et al., 2024; Fasihuddin, 2023; Heinemann & Schroeder, 2023; Hunvik et al., 2021; K. H. A. Tan et al., 2023; Lui et al., 2025; Martins Nunes Avellar et al., 2024; Moersler et al., 2024; Saleh et al., 2021; Sukirman et al., 2024; Sunday et al., 2022; Theethum et al., 2021; Wee et al., 2022)). These visual elements effectively conveyed information, captured users' attention, explained rules, and defined interactive zones within virtual spaces.

Second, virtual environments designed for children or beginner participants typically included more graphical symbols and minimal text (e.g., Klochko et al., 2022; Lu et al., 2022; Segura et al., 2020), as shown in Fig. 5.

This approach was likely intended to foster simple, intuitive interactions that align with younger learners' cognitive, comprehension, and learning capabilities. These environments frequently featured brighter background colors (Klochko et al., 2022; Segura et al., 2020) to maintain engagement. Additionally, a number of studies (Fig. 6) incorporated avatars or virtual tutors that offered engaging, narrative-driven guidance and adaptive feedback to support the learning process (Hui et al., 2024; Martins Nunes Avellar et al., 2024; Parmar et al., 2023).

3.3.3.2. Content-driven interactions patterns. The learning topic mainly influenced the type of interaction, with studies focusing on similar content that frequently employed comparable designs. For example, in three applications (i.e., Akbulut et al., 2018; Alexander et al., 2022;

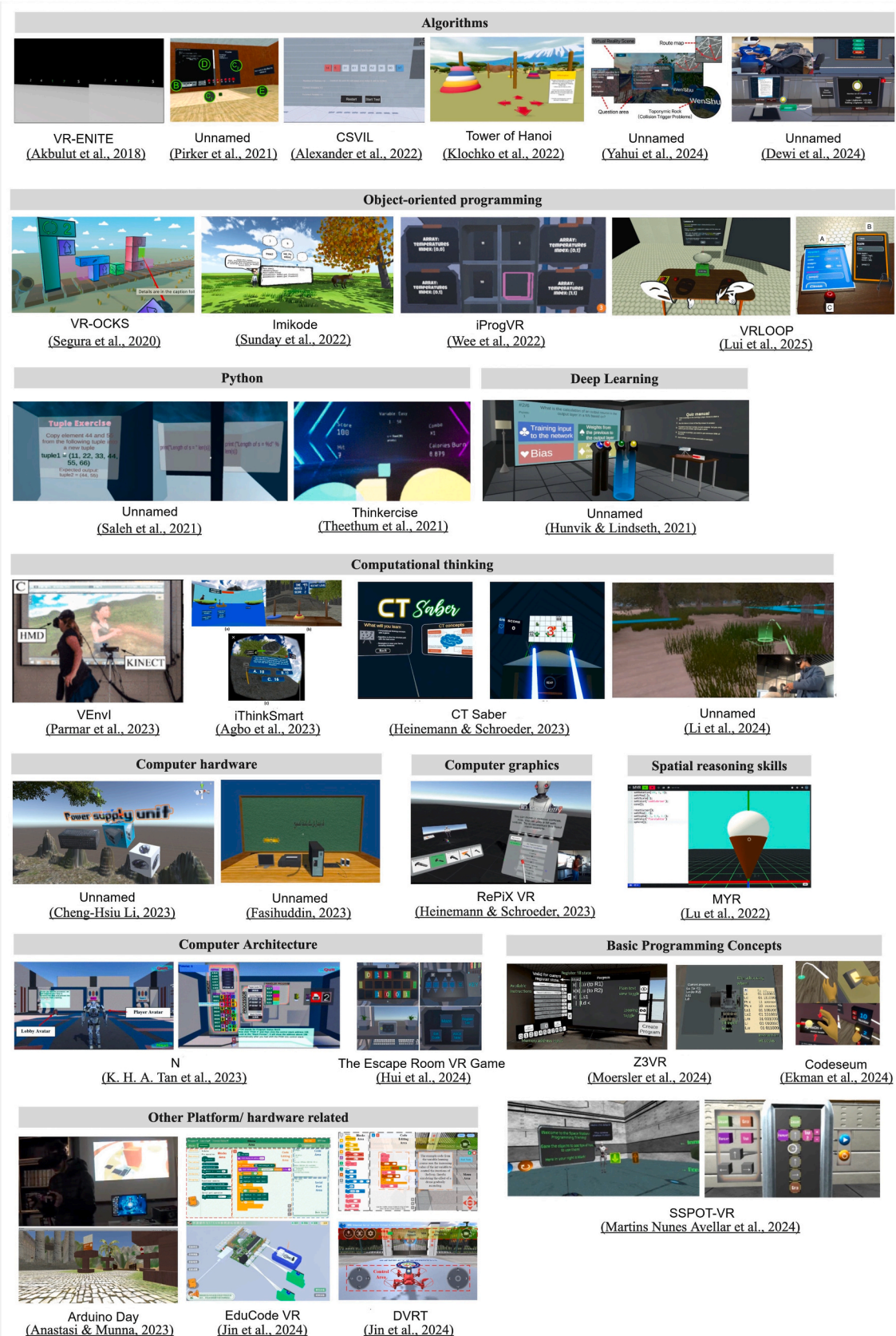


Fig. 4. Key visual presentations of the 29 applications.



Fig. 5. VR applications designed for children and beginner learners, featuring simple graphics, minimal text, and intuitive interactions.

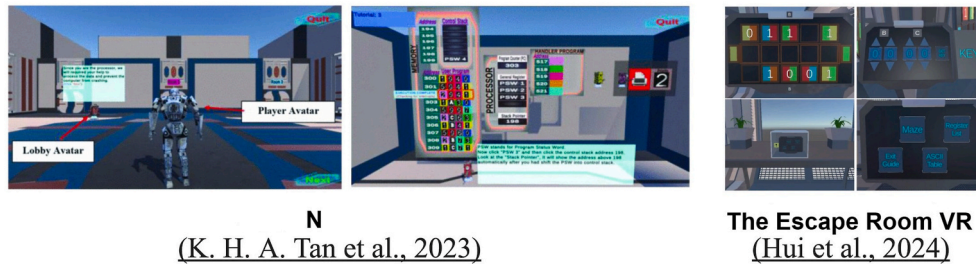


Fig. 6. VR applications using virtual assistants or avatars to provide guidance and feedback during learning tasks.

Pirker et al., 2021) shown in Fig. 7, users can select a specific algorithm and observe the automatic movement of elements to understand its functions in a step-by-step manner. In the Tower of Hanno (Agbo, Olaleye, et al., 2023) and iThinkSmart (Klochko et al., 2022), users were introduced to the task through pre-made instructional videos that demonstrated the correct sequence of moves (Fig. 8).

Similarly, studies focusing on OOP concentrated on basic syntactic concepts such as loops, function calls, and object instantiation through embodied object manipulation (Fig. 9; Heinemann & Schroeder, 2023; Lui et al., 2025; Segura et al., 2020; Sunday et al., 2022). These applications typically included repetitive exercises and explanations to support learning, acknowledging that users are likely to make mistakes during the process. In these cases, learners were required to continue issuing commands and experimenting with new answers as part of the learning experience.

The contextualization of VR tasks for practical learning topics, in which CS concepts are embedded within real-world scenarios, is a growing trend in recent studies (Anastasi & Munna, 2023; Jin, Bai, Song, Yu, & Yue, 2024; Jin, Bai, Song, Yu, & Yue, 2024, Fig. 10). For example, in DVRT, students used graphical programming to control virtual drones for tasks such as irrigation and spraying (Jin, Bai, Song, Yu, & Yue, 2024). These immersive environments utilize VR technology to integrate programming, control logic, and hardware design. By enabling students to interact with abstract CS concepts in realistic, embodied contexts, these applications enhanced motivation and supported the holistic development of STEM (Science, Technology, Engineering, and

Mathematics) skills.

3.3.3.3. Collaborative and multiuser design. With respect to collaboration, the majority of studies emphasized individual interaction within virtual environments but overlooked attention to multiuser or collaborative features (Fig. 11). Among the scenarios reviewed, only two studies explicitly mentioned interaction between teachers and students (Akbulut et al., 2018; Alexander et al., 2022). Notably, Lui et al. (2025), utilizing VRLOOP, were the first to introduce cooperative gameplay for learning object-oriented programming, thereby enabling peers to solve tasks together in a shared virtual space. This feature represents a significant shift from earlier designs and more closely reflects real-life classroom dynamics.

3.4. RQ3. Influence of VR applications on learning outcomes in CS education

This section synthesizes the impact of VR applications on learning outcomes from two perspectives, namely, user performance and user feedback (Table 2).

User performance refers to whether users achieved measurable improvements in learning outcomes. Out of the 29 studies, 21 reported significant improvements based on objective or subjective evidence. Most commonly, researchers assessed performance through objective measures such as quizzes, final exams, or pre- and post-test comparisons (Akbulut et al., 2018; Jin, Bai, Song, Yu, & Yue, 2024; Jin, Bai, Song, Yu,

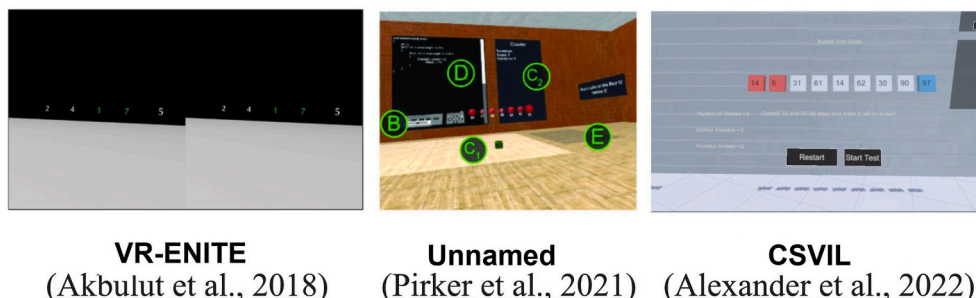
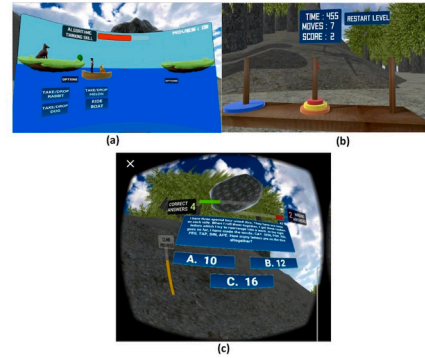


Fig. 7. Similar interactions in algorithm applications: Interactive visualizations of algorithm execution supporting step-by-step understanding.



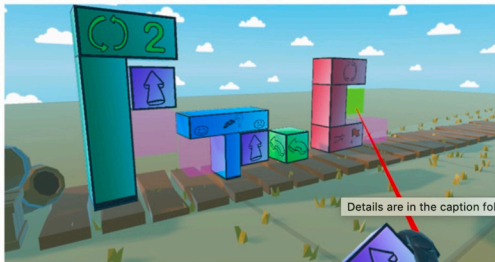
Tower of Hanoi
(Klochko et al., 2022)



iThinkSmart
(Agbo et al., 2023)

Fig. 8. Similar processes in algorithm applications: Instructional sequences demonstrating correct procedures before users begin algorithm tasks.

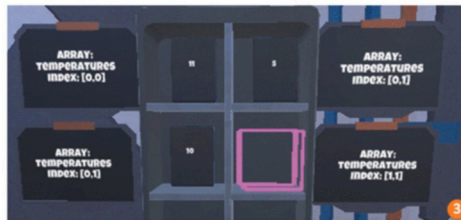
Object-oriented programming



VR-OCKS
(Segura et al., 2020)



Imikode
(Sunday et al., 2022)



iProgVR
(Wee et al., 2022)



VRLOOP
(Lui et al., 2025)

Fig. 9. Similar interactions in OOP-focused applications: Embodied interactions with virtual objects.

& Yue, 2024; Klochko et al., 2022; Theethum et al., 2021). Others used indirect performance indicators, such as task completion rates, number of levels passed, or puzzles solved (Moersler et al., 2024; Segura et al., 2020). A few studies relied on subjective measures, including questionnaires for evaluating perceived effectiveness (Alexander et al., 2022) or self-reported changes in cognitive skills using Likert scales (Jin, Bai, Song, Yu, & Yue, 2024; Li et al., 2024).

Although a number of studies (e.g., Saleh et al., 2021) reported mixed or limited learning gains, their results were generally attributed to challenges in usability or issues in content design and not the limitations of VR as a learning medium. Other studies indirectly evaluated performance, because their focus was on usability, or they served as pilot studies to collect general feedback (Anastasi & Munna, 2023; Heinemann & Schroeder, 2023; Hui et al., 2024; Hunvik et al., 2021; Lui et al., 2025; Pirker et al., 2021; Saleh et al., 2021). In summary, the reviewed studies generally report positive learning outcomes associated

with VR interventions in CS education.

User feedback typically addressed satisfaction, acceptance, and perceptions of system performance, with results reinforcing these dimensions. Although individual studies varied in focus, the majority conducted evaluation by assessing the basic functionality of their prototypes using standardized measures (e.g., system usability scales) and by capturing users' initial perspectives on the overall design. Feedback was subsequently collected through questionnaires and interviews, which informed iterative refinement and broader testing on larger participant groups. Overall, user feedback across studies was overwhelmingly positive. Table 2 indicated that the participants in 26 studies provided positive ratings for VR as a teaching method (excluding Agbo, Olaleye, et al., 2023; Akbulut et al., 2018; Dewi et al., 2024, who did not report relevant feedback). The majority of users provided positive emotional and cognitive responses, noting enjoyment, increased interest, and ease of use. VR was frequently described as fun, motivating,

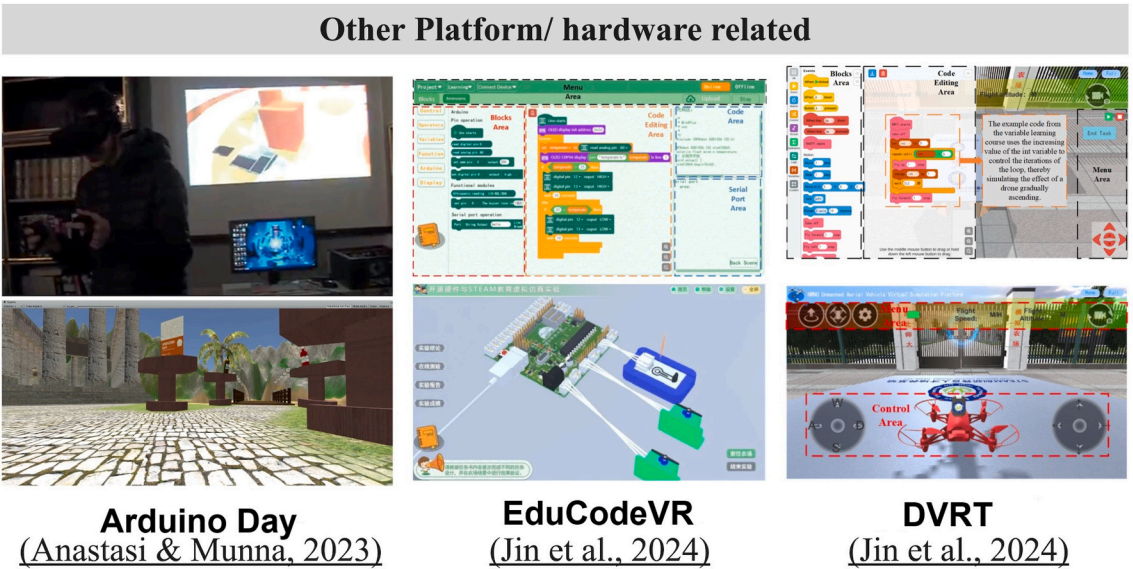


Fig. 10. Contextualized applications: Integrating programming tasks into real-world-inspired VR scenarios.



Fig. 11. Multiuse/collaborative applications.

Table 2
User performance and feedback.

No.	Application Name	Test content	User performance	User feedback
1	VR-ENITE (Akbulut et al., 2018)	Algorithms; multiple-choice test (quiz score)	Average scores (control group: 60; experimental group: 72).	–
2	VR-OCKS (Segura et al., 2020)	Ex1: User interface Ex2: System acceptance Ex3: Programming puzzles	Children using VR-OCKS passed 25% more levels than did children in the control group.	Children perceive the games as engaging and fun. The ratings are mostly more than 4 out of 5.
3	Unnamed (Hunvik et al., 2021)	Ex1: Background, usability, ease of use Ex2: Engagement, user experience)	–	Most users perceive it as fun, easy to understand, and an effective means of learning.
4	Unnamed (Pirker et al., 2021)	Engagement, emotion, and learning	–	Feedback on VR application in all aspects is better than that on web application.
5	Unnamed (Saleh et al., 2021)	Ex1: User perception Ex2: Novice's acceptance	–	Ex1: Average score is 3/5; 73% users suggested improvement in the virtual set. Ex2: 7 out of 10 questions scored 2.5 points or less; dizziness and hand muscle fatigue.
6	Thinkercise (Theethum et al., 2021)	User performance (exam and calorie consumption)	Students who read books on and study with the help of this VR intervention exhibited improved average grades by 40.83%. Students who only read books displayed improvement by 10.83%.	83.33% of the participants provided positive feedback (4.46/5). Users preferred VR learning than book learning.
7	VENVI (Parmar et al., 2023)	User performance (cognition, presence, attitudes, and programming performance)	The students scored significantly higher on all aspects using VR. People in the immersive embodied VR indicated that they gained knowledge.	77% of the students provided positive feedback.
8	CSVIL (Alexander et al., 2022)	User experience (VR familiarity, perception, environment, and decor), learning outcomes	All students approved of the effectiveness of the VR tutorial (12 = very good, 2 = good). Nine users (out of 14) indicated that they were familiar with VR simulations.	13 users perceived that the VR control system is simple. The majority of students gave positive ratings on overall experience (12 out of 14).
9	Imikode (Sunday et al., 2022)	Usefulness, ease of use, and learning, satisfaction	–	Quantitative: more than 80% of the students reported improvements in their programming skills. The average Cronbach's alpha value is 0.85 (high). Qualitative: positive feedback on problem solving.
10	iProgVR (Wee et al., 2022)	Ex1: Feasibility Ex2: Impact of VR interventions (knowledge, outcomes, and mental workload)	Ex2: All participants exhibited higher scores after undergoing VR interventions. The analogy of display in VR is more appealing than verbal narration in class.	Ex1: 43 participants (out of 60) preferred VR intervention (iProgVR) over video lectures.
11	Tower of Hanoi (Klochko et al., 2022)	Grade point average (at the beginning and after teaching)	GPA of students in the experimental group (7.9) is higher than that of control group (6.8); $p < 0.01$.	1. 91.7% of students displayed interest in using VR software in CS courses; 2. 58.3% of students exhibited interest in using this application.
12	MYR (Lu et al., 2022)	Spatial reasoning skills (the score received for correct descriptions of 3D objects)	The post-test score of the Spring group improved; no significant improvement in the scores of the Fall group (that with higher basic skills).	Students rated the entire program as 4 (out of 5).
13	iThinkSmart (Agbo, Olaleye, et al., 2023)	Cognitive test (self-assessment)	Positive effects of game elements and VR learning environments on learners' perceptions.	–
14	Unnamed (Cheng-Hsiu Li, 2023)	Learning satisfaction, learning outcome	Participants performed better on the post-test than on the pre-test (immediate test and the test 5 weeks after the experiment).	Fondness: 4.37/5 Motivation: 4.3/5 Resistance: 4.4/5.
15	Unnamed (Fasihuddin, 2023)	Ease of use, usefulness	–	More than 80% of people strongly believed that this prototype is easy to use. More than 90% of people strongly believed that usability is good.
16	RePiX VR (Heinemann & Schroeder, 2023)	Usability, skill, consequence	–	Results of 10-point Likert-type scale: usability (7.46), skill (7.58). The majority of people provided positive comments.
17	N (K. H. A. Tan et al., 2023)	Learning outcomes, learning experience and preference.	The experiment group achieved a higher overall quiz score (6.5 vs. 5.33). Scores for “Fetching and Executing Instructions” were comparable (3.50 vs. 3.67). For “Interrupt Handling” and “Interrupt-Driven I/O,” the experiment group achieved significantly higher scores (3.00 vs. 1.67).	1. All participants reported enjoying the VR game and a better understanding of computer architecture concepts. They welcomed more serious games for learning. 2. Minor issues included slower robot avatar movement and a certain degree of cybersickness reported in the HMD mode.
18	CT Saber (Sukirman et al., 2024)	Usability	–	1. High usability scores: usefulness (80.12), ease of use (79.92), ease of learning (84.20), satisfaction (78.56) out of 100. 2. Regression analysis indicated that usefulness was the only significant predictor of satisfaction. 3. The majority of open-ended responses were positive—pointing out praises for an interactive and joyful way of learning. The realism and immersion provided by the game were also highlighted.
19	Z3VR (Moersler et al., 2024)	Programming tasks, usability	In the second evaluation for the improved version (n = 15), 80% of participants attempted programming tasks; nearly all successfully completed the tasks, even	SUS score improved significantly from 52.86 (early version) to 70.83 (improved version). Participants found the introduction effective and the system helpful for understanding basic

(continued on next page)

Table 2 (continued)

No.	Application Name	Test content	User performance	User feedback
			those without prior low-level programming experience.	concepts in programming.
20	The Escape-Room VR Game (Hui et al., 2024)	Game design, learning experience, and curiosity companion	–	A number of potential user-friendly improvements: add gamification, improve UI. More than 73% of participants rated 4 or 5 for immersion and engagement, while 60% agreed that the VR game supported their understanding. 13 out of 15 students mentioned that the curiosity companion enhanced immersion, while 14 out of 15 reported that it aided learning.
21	Unnamed (Yahui et al., 2024)	Learning outcomes (Dijkstra's algorithm test (theory + programming exams); teaching effectiveness)	The scores of the VR group scored were higher than those of the traditional group in theory and programming tests.	The survey demonstrated that 78% of the <i>already learned</i> group and 86% of the <i>not yet learned</i> group recognized VR's effectiveness. Most students are recognized by the virtual simulation teaching efficient knowledge transfer ability.
22	Academic Museum of Technology/Arduino Day (Anastasi & Munna, 2023)	User perception (possibility of using AR and VR as vehicles for increased awareness of their knowledge)	–	Students expressed that VR technologies can be viewed as an enthusiastic approach, while a very strong will is required to deepen their curricular use (items were rated using a five-point Likert-type scale, with customized questions).
23	Unnamed (Li et al., 2024)	Computational thinking ability	Significant improvement across all dimensions after VR teaching: creativity (3.55 → 3.99, $p = 0.003$), algorithmic thinking (3.20 → 3.97, $p = 0.000$), collaboration (3.44 → 4.11, $p = 0.000$), critical thinking (3.37 → 4.46, $p = 0.000$), problem solving (3.56 → 4.39, $p = 0.000$).	Students generally reflected that they were more relaxed during the learning process and can better understand and master terrain creation.
24	Codeiseum (Ekman et al., 2024a, 2024b)	Perceived user engagement, system usability	VR led to higher scores in focused attention, reward, and aesthetic appeal (UES-SF median = 4.04 vs. 3.21 for desktop, $p < 0.01$). No significant difference was noted for SUS scores between the VR and desktop groups (72 vs. 67.5, $p = 0.1$).	Participants reported VR as more immersive and fun, although they emphasized physical demands and occasional motion sickness in VR.
25	EduCodeVR (Jin, Bai, Song, Yu, & Yue, 2024)	Learning outcomes (final exam via a graphical programming test), usability, presence, and course engagement	Final exam scores: mean (VR group) = 94.55 vs. control group = 94.21. Improved STEAM literacy (the majority of scores for the four dimensions—creativity, cooperative thinking, problem solving, and STEAM literacy—reached above 4. The score for programming skills is relatively lower, with a median of 3.8 and an overall range below 4.)	1. High immersion and engagement ratings (average > 4/5 on a Likert-type scale). 2. Positive emotional responses; the students reported more enjoyment and involvement in VR. 3. Usability scores consistently high among experts and students. 4. Minor issues with comfort in movement and adaptation of first-time users of VR.
26	SSPOT-VR (Martins Nunes Avellar et al., 2024)	Acceptance, knowledge retention	Average pre-test score: 3.9/8 (48.75%); post-test score: 6.55/8 (81.88%); significant improvement (Wilcoxon signed-rank test: $Z = -4.782$, $p = 0.000002$).	Students reported high levels of acceptance and engagement; average user experience score was 82.79/100 (online) and 82.42/100 (in-class VR group).
27	Unnamed (Dewi et al., 2024)	Learning outcomes	Average pre- and post-test score for control (traditional) group: 53 → 64 and experimental (VR) group: 61 → 87. The VR group displayed a significantly higher improvement in students' score ($p = 0.000$) compared with the traditional group.	–
28	DVRT (Jin, Bai, Song, Yu, & Yue, 2024)	Learning outcomes, usability, game engagement, presence	1. Average score of the experimental group: 94.56, SD: 1.84; control group: 94.14, SD: 1.80 2. Male participants in the VR group outperformed those in the control group ($p = 0.13$); women exhibited no difference ($p = 0.93$).	1. High ratings on immersion, presence, and learning satisfaction; positive emotions (avg. 4.0), low negative emotions (avg. 2.0) 2. Students reported that the course was engaging and challenging; presence and motivation were enhanced overall.
29	VRLOOP (Lui et al., 2025)	Task load, usability, and user perception	–	1. Low workload and high perceived performance. High scores in enjoyment ($M = 6.67$ [collaborative]; $M = 6.46$ [individual]), ease of use ($M = 6.67$ [collaborative]; $M = 5.66$ [individual]), usefulness ($M = 5.97$ [collaborative]; $M = 5.52$ [individual]). 2. Students provided positive feedback (13 out of 14), highlighting that visualization and VR experience rendered content easier to understand.

and more intuitive compared with traditional or desktop-based learning environments (Ekman et al., 2024a, 2024b; Jin, Bai, Song, Yu, & Yue, 2024).

A number of studies also examined specific cognitive dimensions of user experience. For example, DVRT (Jin, Bai, Song, Yu, & Yue, 2024) and VRLOOP (Lui et al., 2025) were used to assess immersion, presence, and workload. These studies reported high levels of immersion and manageable task demands; however, they also identified certain challenges, including motion sickness (Ekman et al., 2024a, 2024b), difficulties in interface adaptation (Saleh et al., 2021), and physical

discomfort (Jin, Bai, Song, Yu, & Yue, 2024). Particularly, in Saleh et al. (2021), the user acceptance score was relatively low (2.5 out of 5), with participants citing dizziness and hand muscle fatigue as significant issues.

Across the selected studies, the majority of applications were implemented on commodity mid-/low-fidelity devices (mobile VR or standalone HMD/PC), with only a few high-fidelity simulator-style environments. While a broad comparison between mobile VR and HMDs has been conducted, both categories exhibited generally positive learning outcomes and user feedback, without a consistent pattern that

links device fidelity to effectiveness and user experience.

Fig. 12(a) depicts that 58.6% of studies ($n = 17$) reported statistically significant learning improvements after VR intervention, while 27.6% ($n = 8$) emphasized learning gains without statistical tests, and only 13.8% ($n = 4$) found no improvement. These proportions indicated that the majority of VR applications yielded measurable or reported learning benefits, thus supporting the current study's conclusion that VR is reported to support conceptual understanding within CS education. Similarly, Fig. 12(b) demonstrates that 89.7% of the studies ($n = 26$) reported positive user feedback, with participants highlighting enjoyment, immersion, and perceived learning effectiveness. Only 10.3% ($n = 3$) provided neutral responses, whereas none reported predominantly negative feedback. This concentration of positive reactions reinforced the motivational advantages of VR observed across the reviewed applications. In summary, despite a few limitations, the overall trend across the studies strongly suggested that VR is an effective and well-received tool for supporting CS education.

4. Discussion

This review synthesizes recent developments in VR applications within CS education and situates them within broader debates in immersive learning and computing education research. While prior reviews have mapped technological trends and gamification features (Agbo, Sanusi, et al., 2021; Lampropoulos & Kinshuk, 2024), the present analysis extends this work by systematically connecting design features, theoretical foundations, and reported learning outcomes.

A dominant pattern across the reviewed studies is VR's effectiveness in supporting conceptual understanding of abstract CS topics such as algorithm behaviour, object instantiation, computational thinking, and hardware architecture. Applications such as VR-ENITE (Akbulut et al., 2018), iProgVR (Wee et al., 2022), and VEnvI (Parmar et al., 2023) demonstrate how immersive visualization and embodied interaction make invisible computational processes observable and manipulable. This finding aligns with Chen's (2009) theoretical argument that VR is particularly suited for domains characterized by abstract or non-tangible phenomena. For example, iProgVR (Wee et al., 2022) employs spatial metaphors such as drawers to represent memory arrays, enabling learners to physically manipulate indices and values. Similarly, VEnvI (Parmar et al., 2023) integrates choreographed movement with computational thinking tasks, operationalizing embodied cognition principles. DVRT (Jin, Bai, Song, Yu, & Yue, 2024) situates programming logic within smart-farm drone simulations, aligning with situated learning theory by embedding abstract code within authentic problem-solving contexts. These findings suggest that VR may support the reduction of abstraction through embodied and spatialized

representations. This can be explained by the alignment between VR's core affordances such as spatial visualization, immersion, and embodied interaction and the nature of conceptual knowledge in computer science. Abstract processes can benefit from being externalized into visual and interactive forms, reducing reliance on purely symbolic reasoning and supporting mental model construction.

However, this strength simultaneously reveals a limitation. When tasks require precise syntactic input, extended debugging, or high-density text processing, immersive environments may increase cognitive load. Mayer's cognitive theory of multimedia learning suggests that excessive visual and interactive elements can overwhelm working memory if not carefully designed. For instance, Saleh et al. (2021) reported usability challenges and dizziness in a Python maze-based VR environment, while Ekman et al. (2024a, 2024b) observed physical fatigue in CodeSeum. Such findings resonate with multimedia learning principles, which caution against excessive sensory load when instructional design is not carefully optimized. This limitation arises because procedural programming tasks demand precision, rapid iteration, and sustained attention to textual detail. These requirements are less compatible with VR interfaces. Such interfaces often involve slower input mechanisms, divided attention, and additional physical and perceptual demands. As a result, VR environments may amplify extraneous cognitive load in tasks that are already cognitively intensive.

Although many applications explicitly reference constructivism or gamification (e.g., Agbo, Olaleye, et al., 2023; Akbulut et al., 2018; Sukirman et al., 2024), deeper theoretical articulation remains uneven. For instance, MYR (Lu et al., 2022) enables learners to construct and share 3D code-driven scenes, reflecting constructivist principles of active knowledge construction. RePiX VR (Heinemann & Schroeder, 2023) integrates avatar-guided tasks grounded in multimedia learning principles, although these theoretical links are not always extensively discussed. Gamification is widespread across applications such as CT Saber (Sukirman et al., 2024), Escape-Room VR (Hui et al., 2024), and Imikode (Sunday et al., 2022). These tools incorporate levels, challenges, and narrative framing to enhance engagement. However, while motivational benefits are frequently reported, fewer studies systematically examine how game mechanics interact with cognitive processing or long-term knowledge retention. This pattern may be explained by the design-driven nature of many VR projects, where development is often led by technological possibilities and user experience considerations rather than pedagogical theory. VR environments require substantial technical expertise and resources, which can shift the focus toward building functional and engaging systems rather than grounding them in well-articulated learning frameworks. In addition, translating established learning theories into immersive and interactive 3D environments is inherently complex, making it difficult for developers to explicitly

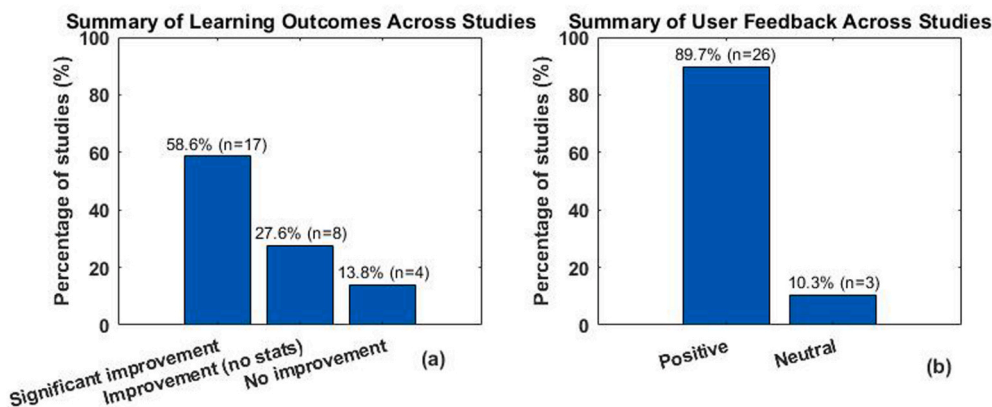


Fig. 12. Summary of learning outcomes and user feedback across the 29 selected articles on VR within CS education. (a) Proportion of studies that reported significant learning improvement, improvement without statistical tests, or no improvement. (b) Proportion of studies that highlighted positive versus neutral user feedback.

operationalise theoretical constructs within VR design. As a result, many applications appear to draw implicitly on familiar concepts such as constructivism or gamification without fully integrating or evaluating them at a theoretical level.

Consequently, engagement features such as gamification are often implemented without sufficient examination of their cognitive and educational impact. As [Chen \(2009\)](#) argues, immersion alone does not guarantee learning effectiveness; pedagogical alignment between interaction design and cognitive processes is essential.

Another significant gap concerns collaboration. Despite programming education's strong grounding in social constructivism and collaborative problem solving, most reviewed applications remain single-user systems. Only a limited number explicitly incorporate shared or cooperative environments, such as CSVIL ([Alexander et al., 2022](#)) and VRLOOP ([Lui et al., 2025](#)). VRLOOP, introduces cooperative object-oriented programming tasks within a shared virtual space. This development may reflect an important shift toward aligning VR affordances with authentic classroom dynamics. The limited presence of collaborative VR environments can be attributed to both technical and design challenges, including synchronization, communication support, and increased system complexity. Additionally, designing meaningful collaborative interactions in immersive environments requires careful integration of pedagogical goals, which may exceed the scope of many current development efforts. As a result, VR applications tend to prioritize individual interaction over socially mediated learning experiences.

Although a majority of studies report significant or perceived learning improvements (e.g., [Dewi et al., 2024](#); [Jin, Bai, Song, Yu, & Yue, 2024](#); [Klochko et al., 2022](#)), evaluations are frequently short-term and limited to pre-post testing or usability surveys. Longitudinal evidence on retention and transfer remains scarce. Additionally, learner diversity is insufficiently explored. DVRT ([Jin, Bai, Song, Yu, & Yue, 2024](#)) reported gender-based differences in outcomes, but few studies systematically examine accessibility, prior programming experience, or cognitive load variation across learner groups. This limitation reflects the early-stage nature of the field, where exploratory prototypes and proof-of-concept studies are prioritised over large-scale, longitudinal investigations. Constraints related to resources, participant recruitment, and the novelty of VR technologies further contribute to the lack of comprehensive and inclusive evaluation designs, limiting the generalisability of current findings.

Overall, the reviewed literature suggests that VR in computer science education is currently more effective as a tool for conceptual augmentation than for supporting full programming practice. This imbalance may reflect the alignment between VR's strengths (visualization, embodiment, immersion) and conceptual learning, alongside its limitations in handling text-intensive, precision-based, and collaborative programming tasks. Addressing these gaps will require a stronger integration of learning theory, interaction design, and empirical evaluation to ensure that VR applications move beyond engaging visualisations toward supporting authentic and inclusive programming learning experiences.

5. Conclusions and future direction

This scoping review synthesized 29 VR applications in CS education and provided an integrated analysis of their technological configurations, design features, theoretical grounding, and reported outcomes. By organizing findings across these interrelated dimensions, this review moves beyond device-oriented mapping and offers a pedagogically informed understanding of how immersive environments are currently being used in CS education.

The evidence indicates that VR demonstrates particular pedagogical value in supporting conceptual understanding of abstract and process-oriented CS topics. Through spatial visualization, embodied

manipulation, and contextualized simulation, immersive environments help externalize invisible computational processes and support mental model development. At the same time, the findings reveal that VR is not universally advantageous. Its strengths are most apparent in conceptual exploration, whereas syntax-intensive programming tasks and extended debugging remain better supported in traditional text-based environments. Thus, VR currently functions most effectively as a complementary instructional medium rather than as a replacement for conventional programming instruction.

More critically, this review identifies a structural gap between technological innovation and theoretical articulation. While many applications implicitly reflect constructivist, embodied, or situated learning principles, explicit theory-driven design remains limited. This fragmentation constrains cumulative knowledge building and reduces the transferability of design insights across contexts. Advancing the field therefore requires a deliberate shift from technology-centered experimentation toward systematically theory-informed development and evaluation.

The review also highlights two underdeveloped areas that will likely shape the next phase of research: collaboration and adaptive intelligence. Despite the inherently social nature of programming practice, most VR applications remain single-user experiences. Multiuser immersive environments are rare, and structured peer interaction is seldom embedded in design. At the same time, recent advances in artificial intelligence create new possibilities for integrating AI-driven virtual peers or tutors within immersive environments. Unlike static avatars or scripted assistants, AI-based agents could provide dynamic scaffolding, real-time feedback, adaptive prompts, and simulated pair-programming dialogue grounded in social constructivist and intelligent tutoring principles. Such systems may help bridge the current gap between immersive visualization and socially mediated learning, particularly in contexts where full networked collaboration is not feasible.

Future research should therefore prioritize four directions: (1) explicit alignment of VR interaction mechanics with established learning theories; (2) expansion into advanced and collaborative CS domains; (3) integration of AI-driven adaptive agents to support socially responsive and personalized learning; and (4) longitudinal and inclusive evaluation frameworks examining retention, transfer, accessibility, and differential learner outcomes.

In conclusion, VR in CS education has progressed from exploratory prototypes to a stage of demonstrated conceptual benefit. However, its maturation as a pedagogically robust medium depends on deeper theoretical grounding, stronger collaborative integration, and rigorous empirical validation. The convergence of immersive VR and AI-driven adaptive systems may represent the next developmental stage of computing education technologies, transforming VR from an engaging enhancement into a strategically designed, socially and cognitively responsive learning ecosystem.

CRedit authorship contribution statement

Azin S. Janani: Writing – review & editing, Writing – original draft, Supervision, Data curation. **Yanzhuo Yang:** Writing – original draft, Data curation. **Xundi Lu:** Writing – original draft, Data curation. **Hai-Ning Liang:** Writing – review & editing. **Nilufar Baghaei:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

Table A3

Publications statistics.

No.	Authors	Year	Application	Topic
1	Akbulut et al. (2018)	2018	VR-ENITE	Sorting algorithms
2	Segura et al. (2020)	2019	VR-OCKS	Basic programming concepts (for loops, while loops, conditional branches)
3	Hunvik and Lindseth (2021)	2021	Unnamed	Deep learning
4	Pirker et al. (2021)	2021	Unnamed	Sorting algorithms (9 types)
5	Saleh et al. (2021)	2021	Unnamed	Python
6	Theethum et al. (2021)	2021	Thinkercise	Python
7	Parmar et al. (2023)	2022	VEnv1	Computational thinking
8	Alexander et al. (2022)	2022	CSVIL	Sorting algorithms, programming paradigms and lecture hall
9	Sunday et al. (2022)	2022	Imikode	Object-Oriented Programming (OOP) concept
10	Wee et al. (2022)	2022	iProgVR	Introductory programming
11	Klochko et al. (2022)	2022	Tower of	Algorithms and programming
12	Lu et al. (2022)	2022	MYR	Spatial reasoning skills
13	Agbo, Olaleye, et al. (2023)	2023	iThinkSmart	Computational thinking
14	Cheng-Hsiu Li (2023)	2023	Unnamed	Computer hardware
15	Fasihuddin (2023)	2023	Unnamed	Computer hardware
16	Heinemann and Schroeder (2023)	2023	RePiX VR	Computer graphics
17	K. H. A. Tan et al. (2023)	2023	Unnamed	Computer Organization and Architecture
18	Sukirman et al. (2024)	2024	CT Saber	Computational thinking
19	Moersler et al. (2024)	2024	Z3VR	Computer Simulation and Basic low-level programming concepts
20	Hui et al. (2024)	2024	Escape-Room VR game	Computer architecture
21	Yang et al. (2024)	2024	Unnamed	Data Structure and Algorithms (DSA)
22	G. Fulantelli et al. (2023)	2023	Academic Museum of Technology; Arduino Day	Application development skills on the Unity platform (Project-Based Learning)
23	Li et al. (2024)	2024	Unnamed	Computational thinking
24	Ekman et al. (2024a, 2024b)	2024	Codeseum	Programming concepts
25	Jin, Bai, Song, Yu, and Yue (2024)	2024	EduCodeVR	Graphical programming and hardware integration (Arduino UNO)
26	Avellar et al. (2024)	2024	SSPOT-VR	Programming (simple programming commands, basic programming control structures)
27	Dewi et al. (2024)	2024	Unnamed	Programming Algorithm
28	Jin, Bai, Song, Yu, Yue, et al. (2024)	2024	DVRT	Graphical programming
29	Lui et al. (2025)	2025	VRLOOP	OOP

References

- Abad-Segura, E., González-Zamar, M.-D., Luque-De La Rosa, A. L. L., & Morales Cevallos, M. B. (2020). Sustainability of educational technologies: An approach to augmented reality research. *Sustainability*, 12(10), 4091. <https://doi.org/10.3390/su12104091>
- Agbo, F. J., Olaleye, S. A., Bower, M., & Oyelere, S. S. (2023a). Examining the relationships between students' perceptions of technology, pedagogy, and cognition: The case of immersive virtual reality mini games to foster computational thinking in higher education. *Smart Learning Environments*, 10(1), 16. <https://doi.org/10.1186/s40561-023-00233-1>
- Agbo, F. J., Oyelere, S. S., Suhonen, J., & Adewumi, S. (2019). A systematic review of computational thinking approach for programming education in higher education institutions. In *Proceedings of the 19th koli calling international conference on computing education research* (pp. 1–10). <https://doi.org/10.1145/3364510.3364521>
- Agbo, F. J., Oyelere, S. S., Suhonen, J., & Laine, T. H. (2021). Co-design of mini games for learning computational thinking in an online environment. *Education and Information Technologies*, 26(5), 5815–5849. <https://doi.org/10.1007/s10639-021-10515-1>
- Agbo, F. J., Oyelere, S. S., Suhonen, J., & Tukiainen, M. (2023b). Design, development, and evaluation of a virtual reality game-based application to support computational thinking. *Educational Technology Research and Development*, 71(2), 505–537. <https://doi.org/10.1007/s11423-022-10161-5>
- Agbo, F. J., Sanusi, I. T., Oyelere, S. S., & Suhonen, J. (2021). Application of virtual reality in computer science education: A systemic review based on bibliometric and content analysis methods. *Education Sciences*, 11(3), 142. <https://doi.org/10.3390/educsc11030142>
- Akbulut, A., Catal, C., & Yildiz, B. (2018). On the effectiveness of virtual reality in the education of software engineering. *Computer Applications in Engineering Education*, 26(4), 918–927. <https://doi.org/10.1002/cae.21935>
- Al-Ansi, A. M., Jaboo, M., Garad, A., & Al-Ansi, A. (2023). Analyzing augmented reality (AR) and virtual reality (VR) recent development in education. *Social Sciences & Humanities Open*, 8(1), Article 100532. <https://doi.org/10.1016/j.ssho.2023.100532>
- Alexander, B., Hou, Y., Khan, B., & Jin, J. (2022). Learn programming in virtual reality? A case study of computer science students. In *IEEE global engineering education conference (EDUCON)* (pp. 270–275). <https://doi.org/10.1109/EDUCON52537.2022.9766621>
- Anastasi, G. F., & Munna, E. G. (2023). Augmented and virtual reality in computer science education. *Communications in Computer and Information Science*, 601–612. https://doi.org/10.1007/978-3-031-29800-4_45
- Arksey, H., & Malley, L. (2005). Scoping studies: Towards a methodological framework. *International Journal of Social Research Methodology*, 8(1), 19–32. <https://doi.org/10.1080/1364557032000119616>
- Berns, C., Chin, G., Savitz, J., Kiesling, J., & Martin, F. (2019). MYR: A web-based platform for teaching coding using VR. In *Proceedings of the 50th ACM technical symposium on computer science education* (pp. 77–83). <https://doi.org/10.1145/3287324.3287482>
- Bouali, N., Nygren, E., Oyelere, S. S., Suhonen, J., & Cavalli-Sforza, V. (2019). Imikode: A VR game to introduce OOP concepts. In *Proceedings of the 19th koli calling international conference on computing education research* (pp. 1–2). <https://doi.org/10.1145/3364510.3366149>
- Chen, C. J. (2009). Theoretical bases for using virtual reality in education. *Themes in Science and Technology Education*, 2, 71–90.
- Cheng-Hsiu, L. (2023). Instructional design, learning satisfaction, and learning outcome in a virtual reality learning environment aimed at improving the cognition of computer hardware components. *International Journal of Engineering and Technology Innovation*, 13(2), 111–124. <https://doi.org/10.46604/ijeti.2023.10247>
- Daily, S. B., Leonard, A. E., Jörg, S., Babu, S., & Gundersen, K. (2014). Dancing alice: Exploring embodied pedagogical strategies for learning computational thinking. In *Proceedings of the 45th ACM technical symposium on computer science education* (pp. 91–96). <https://doi.org/10.1145/2538862.2538917>
- Dengel, A. (2020). How important is immersion for learning in computer science unplugged games?. In *Proceedings of the 51st ACM technical symposium on computer science education* (pp. 1165–1171). <https://doi.org/10.1145/3328778.3366837>
- Dewi, I. P., Ambiyar, A., Effendi, H., Giatman, M., Hanafi, H. F., & Ali, S. K. (2024). The impact of virtual reality on programming algorithm courses on student learning outcomes. *International Journal of Learning, Teaching and Educational Research*, 23(10), 45–61. <https://doi.org/10.26803/ijlter.23.10.3>
- Ekman, J., Solsona, J., & Quintero, L. (2024a). Codeseum: Learning introductory programming concepts through virtual reality puzzles. In *Imx 2024 - Proceedings of the 2024 ACM international conference on interactive media experiences* (Vol. 24, pp. 192–200). <https://doi.org/10.1145/3639701.3656306>
- Ekman, J., Solsona, J., & Quintero, L. (2024b). Codeseum: Learning introductory programming concepts through virtual reality puzzles. In *Proceedings of the 2024 ACM international conference on interactive media experiences* (pp. 192–200). <https://doi.org/10.1145/3639701.3656306>

- Fasihuddin, H. (2023). Virtual reality technology in teaching computer hardware: A prototype and assessment from user perspectives. *TEM Journal*, 899–907. <https://doi.org/10.18421/tem122-36>
- Heinemann, B., Görzen, S., & Schroeder, U. (2022). RePiX VR - Learning environment for the rendering pipeline in virtual reality. *Eurographics 2022 - Education Papers*, 9–16. <https://doi.org/10.2312/EGED.20221040>
- Heinemann, B., Görzen, S., & Schroeder, U. (2023). Teaching the basics of computer graphics in virtual reality. *Computers & Graphics*, 112, 1–12. <https://doi.org/10.1016/j.cag.2023.03.001>
- Heinemann, B., & Schroeder, U. (2023). Evaluating usability and user feedback in an immersive virtual reality environment for computer science education. *Lecture Notes in Computer Science*, 718–724. https://doi.org/10.1007/978-3-031-42682-7_67
- Hui, M. T. R., Liu, S., Qiu, Y., Zhu, G., & Fan, X. (2024). A virtual reality escape room game for learning computer architecture concepts, 794, 385–397 https://doi.org/10.1007/978-3-031-44981-9_32.
- Hunvik, S. R. B., Lindseth, F., Agrati, L. S., Burgos, D., Ducange, P., Limone, P., Perla, L., Picerno, P., & P. (2021). Making use of virtual reality for artificial intelligence education. In *Bridges and mediation in higher distance education* (Vol. 1344, pp. 56–70). Springer International Publishing. https://doi.org/10.1007/978-3-030-67435-9_5.
- Jin, Z., Bai, Y., Song, W., Yu, Q., & Yue, X. (2024). EduCodeVR: VR for programming teaching through simulated farm and traffic. *Visual Computer*. <https://doi.org/10.1007/s00371-024-03699-3>
- Jin, Z., Bai, Y., Song, W., Yu, Q., Yue, X., & Jia, X. (2024). DVRT: Design and evaluation of a virtual reality drone programming teaching system. *Computers & Graphics*, 125, Article 104114. <https://doi.org/10.1016/j.cag.2024.104114>
- Klocho, O. V., Tkachenko, S. V., Babiichuk, I. M., Fedorets, V. M., Galych, T. V., Ignatenko, V., Kharchenko, V., Kobets, H., Kravtsov, Y., Tarasich, V., Ermolayev, D., & Esteban, V. (2022). Using virtual reality technologies for teaching computer science at secondary school. In *ICTERI 2021 workshops* (Vol. 1635, pp. 231–246). Springer International Publishing. https://doi.org/10.1007/978-3-031-14841-5_15.
- Lampropoulos, G., & Kinshuk. (2024). Virtual reality and gamification in education: A systematic review. *Educational Technology Research and Development*, 72(3), 1691–1785. <https://doi.org/10.1007/s11423-024-10351-3>
- Leonard, A. E., Daily, S. B., Jörg, S., & Babu, S. V. (2021). Coding moves: Design and research of teaching computational thinking through dance choreography and virtual interactions. *Journal of Research on Technology in Education*, 53(2), 159–177. <https://doi.org/10.1080/15391523.2020.1760754>
- Li, X., Zhao, M., Ai, D., Zhou, X., & Xiang, H. (2024). An empirical study on the design-based STEM + VR teaching approach to cultivate college students' computational thinking. In *Proceedings of the 2024 9th international conference on distance education and learning* (pp. 217–222). <https://doi.org/10.1145/3675812.3675867>
- Lin, L., Parmar, D., Babu, S. V., Leonard, A. E., Daily, S. B., & Jörg, S. (2017). How character customization affects learning in computational thinking. In *Proceedings of the ACM symposium on applied perception* (pp. 1–8). <https://doi.org/10.1145/3119881.3119884>
- Lu, J., Seavey, L., Zuk, S., Dimino, J., & Martin, F. (2022). Evaluating student spatial skills learning in a virtual reality programming environment. In *IEEE frontiers in education conference (FIE)* (pp. 1–9). <https://doi.org/10.1109/FIE56618.2022.9962403>
- Lui, L. F., Frau, V., Makransky, G., Koumaditis, K., & Leiva, G. (2025). VRLOOP: Immersive virtual reality for learning object-oriented programming. In *2025 IEEE conference on virtual reality and 3D user interfaces abstracts and workshops (VRW)* (pp. 663–670). <https://doi.org/10.1109/VRW66409.2025.00137>
- Moersler, L., Plecher, D. A., Eichhorn, C., & Klinker, G. (2024). Z3VR - An interactive VR learning environment for one of the first computers (konrad zuse's Z3). In *Virtual, augmented and mixed reality* (pp. 123–141). Springer. https://doi.org/10.1007/978-3-031-61047-9_8.
- Nunes Avellar, M., Fioravanti, G., Simao De Deus, M. L., Lucas Jaquie Castelo, W., Branco, K. R., & Barbosa, E. F. (2024). SSPOT-VR: An immersive and affordable mobile application for supporting K-12 students in learning programming concepts. *Education and Information Technologies*, 29(13), 16411–16439. <https://doi.org/10.1007/s10639-024-12499-0>
- Parmar, D., Isaac, J., Babu, S. V., Souza, N., Leonard, A. E., Jorg, S., Gundersen, K., & Daily, S. B. (2016). Programming moves: Design and evaluation of applying embodied interaction in virtual environments to enhance computational thinking in middle school students. *IEEE Virtual Reality*, 131–140. <https://doi.org/10.1109/VR.2016.7504696>
- Parmar, D., Lin, L., Dsouza, N., Jörg, S., Leonard, A. E., Daily, S. B., & Babu, S. V. (2023). How immersion and self-avatars in VR affect learning programming and computational thinking in middle school education. *IEEE Transactions on Visualization and Computer Graphics*, 29(8), 3698–3713. <https://doi.org/10.1109/TVCG.2023.3169426>
- Pirker, J., Dengel, A., Holly, M., & Safikhani, S. (2020). Virtual reality in computer science education: A systematic review. In *Proceedings of the 26th ACM symposium on virtual reality software and technology* (pp. 1–8). <https://doi.org/10.1145/3385956.3418947>
- Pirker, J., Kopf, J., Kainz, A., Dengel, A., & Buchbauer, B. (2021). The potential of virtual reality for computer science education -engaging students through immersive visualizations. In *2021 IEEE conference on virtual reality and 3D user interfaces abstracts and workshops (VRW)* (pp. 297–302). <https://doi.org/10.1109/VRW52623.2021.00060>
- Saleh, A. Y., Chin, G. S., Tei, R., Othman, M. K., Mohamad, F. S., & Chen, C. J. (2021). Learning scope of python coding using immersive virtual reality. In *Innovative systems for intelligent health informatics* (Vol. 72, pp. 1086–1100). Springer International Publishing. https://doi.org/10.1007/978-3-030-70713-2_97
- Segura, R. J., Del Pino, F. J., Ogáyar, C. J., & Rueda, A. J. (2020). VR-OCKS: A virtual reality game for learning the basic concepts of programming. *Computer Applications in Engineering Education*, 28(1), 31–41. <https://doi.org/10.1002/cae.22172>
- Sukirman, S., Ibarahim, L. F. M., Said, C. S., & Murtiyasa, B. (2024). Development and usability testing of a virtual reality game for learning computational thinking. *International Journal of Serious Games*, 11(3), 19–43. <https://doi.org/10.17083/ijsg.v11i3.670>
- Sunday, K., Ocheja, P., Hussain, S., Oyeler, S. S., Samson, B. O., & Agbo, F. J. (2020). Analyzing student performance in programming education using classification techniques. *International Journal of Emerging Technologies in Learning (ijet)*, 15(2), 127. <https://doi.org/10.3991/ijet.v15i02.11527>
- Sunday, K., Oyeler, S. S., Agbo, F. J., Aliyu, M. B., Balogun, O. S., & Bouali, N. (2023). Usability evaluation of imikode virtual reality game to facilitate learning of object-oriented programming. *Technology, Knowledge and Learning*, 28(4), 1871–1902. <https://doi.org/10.1007/s10758-022-09634-6>
- Sunday, K., Wong, S. Y., Samson, B. O., & Sanusi, I. T. (2022). Investigating the effect of imikode virtual reality game in enhancing object oriented programming concepts among university students in Nigeria. *Education and Information Technologies*, 27(5), 6819–6845. <https://doi.org/10.1007/s10639-022-10886-z>
- Tan, K. H. A., Cai, Y., Lokesh, B. T., Zhu, Y., Su, C., & Cao, Q. (2023). VR serious game for learning the computer Organisation and architecture course. In *2023 3rd international conference on educational technology (ICET)* (pp. 1–6). <https://doi.org/10.1109/icet59358.2023.10424233>
- Theethum, T., Arpornrat, A., & Vittayakorn, S. (2021). Thinkercise: An educational VR game for python programming. In *18th international conference on electrical Engineering/electronics, computer, telecommunications and information technology (ECTI-CON)* (pp. 439–442). <https://doi.org/10.1109/ECTI-CON51831.2021.9454730>
- Wee, C., & Yap, K. M. (2021). Design and analysis of a virtual reality game to address issues in introductory programming learning. In *Intelligent technologies for interactive entertainment*, 377 pp. 243–254. Springer International Publishing. https://doi.org/10.1007/978-3-030-76426-5_16.
- Wee, C., Yap, K. M., & Lim, W. N. (2022). iProgVR: Design of a virtual reality environment to improve introductory programming learning. *IEEE Access*, 10, 100054–100078. <https://doi.org/10.1109/ACCESS.2022.3204392>
- Yahui, Y., Yanxia, G., Ling, P., Yingxi, X., Huan, H., & Yiwen, L. (2024). Propelling pedagogical paradigms: A study of advanced human-computer interaction learning in data structure and algorithms based on virtual reality technology. *Communications in Computer and Information Science*, 220–233. https://doi.org/10.1007/978-981-97-0737-9_21